

Olympiad Test Paper — Maths (Class 7) — Paper 4

Chapter 14: Constructions and Tilings

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	14 — Constructions and Tilings
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Treating 'close to 360° ' or 'under 180° ' as enough to tile, copying the bisection trick onto trisection, forgetting the triangle inequality and angle-sum feasibility before constructing, and confusing perpendicular bisectors (equidistant from vertices) with angle bisectors (equidistant from sides). Do not write on this paper — mark answers on your answer sheet.

1. Three towns A, B and C are not on one line. A water tank must be built equidistant from all three towns. Using only the perpendicular-bisector construction, where is the unique spot?

- (A) At the crossing of the perpendicular bisectors of AB and of BC; that point is equally far from A, B and C.
- (B) At the crossing of the three angle bisectors of triangle ABC, which is equally far from the three towns.
- (C) At the midpoint of the longest of the three connecting segments.
- (D) Anywhere on the perpendicular bisector of AB, since that line is equidistant from A and B.

2. A segment PQ is 10 cm long. A point R is built so that $RP = RQ = 6$ cm by swinging equal arcs from P and Q. How far is R from the midpoint M of PQ?

- (A) 1 cm, since $6 - 5 = 1$.
- (B) 6 cm, since R is built with radius 6 cm.
- (C) $\sqrt{11}$ cm, since RM is perpendicular to PQ and $RM^2 = 6^2 - 5^2 = 11$.
- (D) $\sqrt{61}$ cm, since $RM^2 = 6^2 + 5^2 = 61$.

3. You drop a perpendicular from an external point F to a line m, getting foot N. You then mark any other point K on m ($K \neq N$). Comparing FN and FK:

- (A) $FN = FK$, because both are distances from the same point F to the same line.
- (B) $FK < FN$, because the perpendicular is the longest such segment.
- (C) FN is the shortest of all distances from F to points of m, so $FN < FK$.
- (D) They cannot be compared without knowing the length of segment NK.

4. A circle is drawn through the two endpoints of a 12 cm chord AB, and its centre O lies 8 cm from the chord (measured perpendicularly). The radius of the circle is:

- (A) 8 cm, the perpendicular distance from O to AB.
- (B) 14 cm, since $6 + 8 = 14$.
- (C) $\sqrt{28}$ cm, since $\text{radius}^2 = 8^2 - 6^2 = 28$.
- (D) 10 cm, since O is on the perpendicular bisector of AB and $\text{radius}^2 = 6^2 + 8^2 = 100$.

5. Two perpendicular bisectors of the three sides of a triangle are constructed and meet at point W. A classmate adds the perpendicular bisector of the third side. It will:

- (A) pass through W as well, because W is already equidistant from all three vertices.
- (B) miss W, since three lines almost never meet at one point.
- (C) pass through W only if the triangle is equilateral.
- (D) meet the other two at the midpoint of the third side.

6. A perpendicular bisector of AB is drawn. A point T is chosen on it with $TA = 13$ cm, and $AB = 24$ cm. The distance from T to the line AB (along the bisector) is:

- (A) 13 cm, since $TA = TB = 13$.
- (B) 1 cm, since $13 - 12 = 1$.
- (C) 5 cm, since the bisector meets AB at the midpoint, giving a right triangle with hypotenuse 13 and leg 12.
- (D) $\sqrt{313}$ cm, since the $\text{distance}^2 = 13^2 + 12^2$.

7. On line m, two points L and R are marked with A as their midpoint ($AL = AR$). Equal arcs from L and R meet at T above m. A student then claims line AT is the perpendicular bisector of LR AND makes a 90° angle with m. Which is correct?

- (A) Both claims are correct: T is equidistant from L and R, so AT is LR's perpendicular bisector through its midpoint A, hence perpendicular to m.
- (B) Only the first is correct; AT need not be perpendicular to m.
- (C) Only the second is correct; AT is perpendicular but not a bisector of LR.
- (D) Neither is correct unless the arcs use radius equal to AL.

8. A 60° angle is constructed, bisected, and the resulting half is bisected again. This smallest angle is then placed beside the original 60° . The combined angle measures:

- (A) 75° , since the smallest piece is $60 \div 4 = 15^\circ$ and $60^\circ + 15^\circ = 75^\circ$.
- (B) 67.5° , since the smallest piece is 7.5° and $60 + 7.5 = 67.5$.
- (C) 90° , since $60^\circ + 30^\circ = 90^\circ$.
- (D) 80° , since the smallest piece is 20° and $60 + 20 = 80$.

9. Which angle below CANNOT be constructed with only a compass and straightedge?

- (A) 75° , as $60^\circ + \text{half of a } 30^\circ \text{ angle } (15^\circ)$.
- (B) 50° , since it would require trisecting a constructible 150° angle, and general trisection is impossible.
- (C) 105° , as $60^\circ + 45^\circ$ (45° being half of 90°).
- (D) 7.5° , by bisecting 60° three times.

10. To construct a 67.5° angle with compass and straightedge, a valid route is:

- (A) trisect a 202.5° angle to get 67.5° directly.
- (B) construct a right angle (90°), bisect to 45° , bisect to 22.5° , then add $45^\circ + 22.5^\circ$.
- (C) bisect a 135° angle, since $135 \div 2 = 67.5$, where 135° is got by trisecting 405° .
- (D) construct 60° and add 7.5° , where 7.5° comes from trisecting 22.5° .

11. A right angle is bisected, the 45° is bisected, and the 22.5° is bisected. The smallest angle obtained is:

- (A) 7.5° , since $90 \div 12 = 7.5$.
- (B) 15° , since two halvings give 22.5 and then $22.5 - 7.5 = 15$.
- (C) 11.25° , since $90 \rightarrow 45 \rightarrow 22.5 \rightarrow 11.25$ over three halvings.
- (D) 5.625° , after four halvings of 90° .

12. You construct a 30° angle (half of 60°) and lay three of them side by side sharing arms, then bisect the whole result once. The final angle is:

- (A) 45° , since three 30° angles make 90° and half of 90° is 45° .
- (B) 30° , since bisecting returns to one 30° piece.
- (C) 60° , since three 30° make 90° and $90 - 30 = 60$.
- (D) 135° , since three 30° make 90° and you add a 45° .

13. An angle of measure 96° is given. A student wants a 24° angle from it using only bisections. Is this possible, and how?

- (A) No, because 24° is not a multiple of 15° and so is never constructible.
- (B) Yes, by trisecting 96° once, since $96 \div 3 = 32$, then bisecting to ... no, it is impossible.
- (C) Yes: bisect 96° to 48° , then bisect 48° to 24° (two halvings, $96 \div 4 = 24$).
- (D) Yes, by bisecting 96° once, since $96 \div 2 = 48$ and 48 is close enough to 24 .

14. To copy an angle ABC of unknown size onto a fresh ray, you swing an arc of radius r across both arms of ABC, cutting them at P and Q, and step off the chord PQ on a same-radius arc at the new vertex. If you accidentally use a SMALLER radius r' at the new vertex but the SAME chord length PQ, the copied angle will be:

- (A) exactly equal, because the chord PQ was copied faithfully.
- (B) larger than the original, because the same chord across a smaller arc subtends a wider angle.
- (C) smaller than the original, because a smaller radius means a smaller angle.
- (D) impossible to predict without measuring the original angle.

- 15.** A segment of length 7 cm is to be copied. Your compass can open to at most 5 cm. Can the segment be copied exactly with compass and straightedge?
- (A) No directly in one step, but yes by stepping off in parts (e.g. 4 cm then 3 cm along a line), since the compass carries each part exactly.
 - (B) No, a segment longer than the compass span can never be copied.
 - (C) Yes, in a single arc, since the compass span does not limit copying.
 - (D) No, because copying a length requires a marked ruler when the compass is too small.
- 16.** A triangle has two sides 9 cm and 4 cm. For how many INTEGER values of the third side (in whole cm) does a genuine triangle exist?
- (A) 7, since the third side x satisfies $5 < x < 13$, giving $x = 6, 7, 8, 9, 10, 11, 12$.
 - (B) 9, since x can be 5, 6, 7, 8, 9, 10, 11, 12, 13.
 - (C) 5, since x ranges over $9 - 4 = 5$ values.
 - (D) 8, since x can be any whole number from 5 to 12.
- 17.** A triangle is to be built by ASA with angles 95° and 86° on a 6 cm base. Before drawing you check feasibility. The triangle:
- (A) cannot be built, since $95^\circ + 86^\circ = 181^\circ$ exceeds 180° , leaving no room for a third positive angle.
 - (B) can be built; the third angle is $180 - 181 = -1^\circ$, which is allowed for an obtuse triangle.
 - (C) can be built, since the base 6 cm is positive and both angles are under 90° together count as acute.
 - (D) cannot be built, because ASA requires the two base angles to be equal.
- 18.** Using SAS you are given sides 6 cm and 8 cm with included angle 90° . The third side (opposite the right angle) comes out as:
- (A) 14 cm, since $6 + 8 = 14$.
 - (B) 2 cm, since $8 - 6 = 2$.
 - (C) $\sqrt{28}$ cm, since $\sqrt{(8^2 - 6^2)} = \sqrt{28}$.
 - (D) 10 cm, since with a right included angle the third side is $\sqrt{(6^2 + 8^2)} = \sqrt{100}$.
- 19.** Which single data set determines EXACTLY ONE triangle (up to congruence)?
- (A) Two sides 7 cm and 9 cm with a 40° angle opposite the 7 cm side.
 - (B) Three angles 40° , 60° and 80° .
 - (C) Two sides 7 cm and 9 cm with the 40° angle lying between them.
 - (D) Two sides 5 cm and 6 cm and the third side 11 cm.
- 20.** A right triangle is built by RHS with hypotenuse 17 cm and one leg 8 cm. The perimeter of the triangle is:
- (A) 42 cm, since the other leg is $17 - 8 = 9$, and $8 + 9 + 17 = 34$... rounded to 42.
 - (B) 50 cm, since the other leg is $17 + 8 = 25$, and $8 + 25 + 17 = 50$.
 - (C) 40 cm, since the other leg is $\sqrt{(17^2 - 8^2)} = 15$, and $8 + 15 + 17 = 40$.
 - (D) It cannot be built, since the leg must equal the hypotenuse.

21. A triangle ABC is built by SAS with $AB = 6$ cm, $AC = 6$ cm and included angle $A = 60^\circ$. The triangle turns out to be:

- (A) right-angled at A, with $BC = 6\sqrt{2}$ cm.
- (B) isosceles but not equilateral, with BC longer than 6 cm.
- (C) impossible, since two sides cannot be equal in SAS.
- (D) equilateral, with $BC = 6$ cm.

22. Two sides of a triangle measure 5 cm and 12 cm and the angle between them is increased steadily from 30° to 150° . The third side (opposite that angle):

- (A) stays constant, since the two given sides are unchanged.
- (B) increases throughout, growing as the included angle widens.
- (C) decreases throughout, since a wider angle folds the ends closer.
- (D) first increases then decreases, peaking at 90° .

23. You are told a triangle has sides 3 cm, 4 cm, and an angle of 90° opposite the longest side. The longest side must be:

- (A) 7 cm, the sum $3 + 4$.
- (B) 5 cm, since with the right angle opposite it, it is the hypotenuse $\sqrt{3^2 + 4^2}$.
- (C) 1 cm, the difference $4 - 3$.
- (D) Any length more than 4 cm, since the right angle does not fix it.

24. A regular polygon has an interior angle of 168° . How many sides does it have?

- (A) 12, since $168 \div 14 = 12$.
- (B) 15, since the exterior angle 12° and $180 \div 12 = 15$.
- (C) 30, since the exterior angle is $180 - 168 = 12^\circ$ and $360 \div 12 = 30$.
- (D) 21, since $168 \div 8 = 21$.

25. Two regular polygons have interior angles differing by 30° . One is a regular hexagon (120°). The OTHER, if it has MORE sides than the hexagon, has interior angle 150° and is a:

- (A) regular dodecagon (12 sides), since exterior angle $= 180 - 150 = 30^\circ$ and $360 \div 30 = 12$.
- (B) regular octagon, since 150° is the octagon's interior angle.
- (C) regular decagon, since a 30° difference means 10 sides.
- (D) regular nonagon, since 150° gives $360 \div 150 = 2.4 \approx 9$ sides.

26. A regular polygon's exterior angle is exactly one-fifth of its interior angle. The polygon is a:

- (A) regular hexagon, since one-fifth points to 6 sides.
- (B) regular pentagon, since the word 'fifth' means 5 sides.
- (C) regular dodecagon (12 sides), since interior $= 5 \times$ exterior and interior + exterior $= 180$ give exterior 30° , so $n = 360 \div 30 = 12$.
- (D) regular decagon, since exterior 36° and $36 = \text{interior} \div 5$.

- 27.** The interior angle of a regular n -gon is $180^\circ - 360^\circ/n$. For which n is the interior angle EXACTLY twice the exterior angle?
- (A) $n = 4$, since a square's angle is double something.
 - (B) $n = 3$, since the triangle's 60° is twice 30° .
 - (C) $n = 12$, since 12 is double 6.
 - (D) $n = 6$, since interior = $2 \times$ exterior gives exterior = 60° and $360 \div 60 = 6$.
- 28.** A regular polygon is such that 360° divided by its interior angle equals 2.5. Although it cannot self-tile, what is its interior angle and number of sides?
- (A) Interior 150° , a regular dodecagon (12 sides).
 - (B) Interior 144° , a regular decagon (10 sides).
 - (C) Interior 135° , a regular octagon (8 sides).
 - (D) Interior 90° , a square (4 sides).
- 29.** Among regular polygons, exactly three self-tile. A student lists their EXTERIOR angles. Which list is correct?
- (A) 120° , 90° , 60° (triangle, square, hexagon).
 - (B) 60° , 90° , 120° (triangle, square, hexagon), reading interior angles by mistake.
 - (C) 72° , 60° , 45° (pentagon, hexagon, octagon).
 - (D) 120° , 90° , 72° (triangle, square, pentagon).
- 30.** Why does the self-tiling ability of regular polygons stop after the hexagon, never resuming for any larger number of sides?
- (A) Because for $n > 6$ the interior angle exceeds 180° , which no tiling allows.
 - (B) Because large polygons have too many sides to fit any vertex.
 - (C) For $n > 6$ the interior angle is strictly between 120° and 180° , so $360 \div (\text{interior})$ lies strictly between 2 and 3 and is never a whole number.
 - (D) Because beyond six sides the exterior angle becomes negative.
- 31.** A regular polygon tiles the plane by itself with exactly k copies at each vertex. The only possible values of k are:
- (A) 3, 4, 5 and 6, covering pentagons too.
 - (B) 2, 3 and 4 only.
 - (C) 3, 4 and 6 (hexagon, square, triangle).
 - (D) any $k \geq 3$, since at least three tiles always meet.
- 32.** A regular polygon has interior angle 150° . A student says 'since 150° is under 180° and $360 \div 150 = 2.4$ rounds to 2, two of them tile.' The correct verdict is:
- (A) It does not self-tile; two give 300° (a 60° gap) and three give 450° (overlap), and rounding the divisor never produces a real tiling.
 - (B) It tiles with two corners, since 2.4 rounds down to 2.
 - (C) It tiles with three corners, since 2.4 rounds up to 3.
 - (D) It tiles, because 150° is under 180° .
- 33.** Of the regular pentagon (108°), octagon (135°) and decagon (144°), how many self-tile the plane?

- (A) Exactly one (the octagon), since 135° is a multiple of 45° .
- (B) Exactly two (octagon and decagon), since both exceed 120° .
- (C) All three, since each interior angle is under 180° .
- (D) None of them, since $360 \div 108$, $360 \div 135$ and $360 \div 144$ are 3.33, 2.67 and 2.5 — none whole.

34. A regular polygon has interior angle A with $90^\circ < A < 120^\circ$. Can such a polygon EVER self-tile?

- (A) Yes, if A is a multiple of 30° , such as the pentagon's 108° .
- (B) No, because $360 \div A$ then lies strictly between 3 and 4, so it is never a whole number.
- (C) Yes, for any A in that range, since three or four corners can fit.
- (D) No, because A in that range exceeds 180° .

35. At a tiling vertex the corner angles are found to total 354° . The correct interpretation is:

- (A) The vertex does not close — there is a 6° gap, so it is not a valid tiling vertex.
- (B) It overlaps by 6° .
- (C) It is valid, since 354° is within rounding distance of 360° .
- (D) The measurement is impossible, because vertex angles can never be under 360° .

36. At a tiling vertex four equilateral triangles (60°) and two squares (90°) are placed. The angle total tells you:

- (A) the vertex closes exactly, since six tiles always fit.
- (B) there is a 60° gap.
- (C) the vertex overlaps by 60° , since $4 \times 60 + 2 \times 90 = 240 + 180 = 420^\circ > 360^\circ$.
- (D) it is valid, since both 60° and 90° divide 360° .

37. A proposed semi-regular vertex is square (90°), square (90°), triangle (60°), triangle (60°). What happens at this vertex?

- (A) It closes exactly, since the angles total 360° .
- (B) It overlaps by 60° , since the angles total 420° .
- (C) It closes exactly, because four tiles are used.
- (D) It does not close: the angles total 300° , leaving a 60° gap.

38. Three tiles meet at a vertex: a regular hexagon (120°), a square (90°), and one more regular polygon that closes the vertex exactly. That third polygon is a:

- (A) regular hexagon (120°), since $360 - 120 - 90 = 150 \approx 120$.
- (B) square (90°), since two squares and a hexagon often tile.
- (C) regular dodecagon (150°), since $360 - 120 - 90 = 150^\circ$.
- (D) equilateral triangle (60°), since the leftover rounds to 60° .

39. At a vertex, one regular dodecagon (150°) and one regular hexagon (120°) are placed, leaving room for ONE more regular polygon. That polygon must be a:

- (A) regular triangle (60°), since $360 - 150 - 120 = 90$ rounds to 60.
- (B) regular hexagon (120°), since hexagons fill leftover space.
- (C) regular polygon of interior angle 90° , but no such polygon exists.
- (D) regular polygon of interior angle 90° , namely a square, since $360 - 150 - 120 = 90^\circ$.

40. A semi-regular vertex uses one regular octagon (135°), one square (90°), and one more regular polygon. The third tile's interior angle, and whether it exists, are:

- (A) 90° , another square, since $360 - 135 - 90 = 135 \approx 90$.
- (B) 135° , another regular octagon, since $360 - 135 - 90 = 135^\circ$ (the 4.8.8 vertex).
- (C) 120° , a regular hexagon, since hexagons close most vertices.
- (D) 135° , but no regular polygon has that interior angle.

41. A vertex is built from one regular dodecagon (150°) and the rest filled with equilateral triangles (60°) only. How many triangles are needed to close it exactly?

- (A) Three triangles, since $360 - 150 = 210$ and $210 \div 60 \approx 3$.
- (B) Four triangles, since $4 \times 60 = 240$ fills the rest.
- (C) Three triangles, giving exactly 360° .
- (D) Three and a half, which is not a whole number, so a dodecagon plus triangles ALONE cannot close this vertex.

42. A semi-regular vertex has one regular hexagon (120°) and the rest filled with equilateral triangles (60°). What is the only whole number of triangles that closes it?

- (A) Two triangles, since $360 - 120 = 240$ and $240 \div 120 = 2$.
- (B) Three triangles, since $3 \times 60 = 180$ looks like enough.
- (C) Four triangles, since $360 - 120 = 240$ and $240 \div 60 = 4$.
- (D) Six triangles, since six triangles always meet at a point.

43. Two regular hexagons (120° each) meet at a vertex, and the rest is to be filled with squares (90°) only. Can it be closed exactly?

- (A) Yes, with one square, since 120 is close to 90.
- (B) No, since $360 - 240 = 120$ and $120 \div 90 = 1.33$ is not a whole number.
- (C) Yes, with two squares, since $2 \times 90 = 180$ fills the rest.
- (D) Yes, with 1.33 squares.

44. A semi-regular vertex uses two equilateral triangles (60°), one square (90°), and one more regular polygon. The fourth polygon's interior angle is:

- (A) 120° , a regular hexagon, since $360 - 60 - 60 - 90 = 150 \approx 120$.
- (B) 90° , another square, since two squares balance two triangles.
- (C) 150° , a regular dodecagon, since $360 - 60 - 60 - 90 = 150^\circ$.
- (D) 150° , but no regular polygon has that interior angle.

45. A vertex uses one square (90°) and three equilateral triangles (60°). What happens with exactly those four tiles?

- (A) It does not close: the angles total 270° , leaving a 90° gap.
- (B) It closes exactly, since the angles total 360° .
- (C) It overlaps by 90° , since the angles total 450° .
- (D) It closes exactly, because four tiles are used.

46. At a tiling vertex one regular pentagon (108°) is placed, and the rest is filled with one regular polygon repeated. For the vertex to close, that polygon's interior angle would have to divide $360 - 108 = 252^\circ$. The smallest such whole-number-count fit needs interior angle:

- (A) 126° , since one tile of 126° plus the pentagon makes 360 ... but $360 - 108 = 252$, not 126 .
- (B) no regular polygon works, since $252 \div 60 = 4.2$, $252 \div 90 = 2.8$, $252 \div 120 = 2.1$ — none whole, and 252 itself is not a regular interior angle.
- (C) 60° , since $252 \div 60 \approx 4$.
- (D) 90° , since $252 \div 90 \approx 3$.

47. Equilateral triangles tile the plane. Does EVERY triangle (scalene, obtuse, any shape) tile the plane?

- (A) No — only the equilateral triangle tiles; others leave gaps.
- (B) Yes — two copies of any triangle joined along a side make a parallelogram, and parallelograms tile by sliding.
- (C) Only acute triangles tile; obtuse ones overlap.
- (D) Only if the triangle's angles each divide 360° evenly.

48. A quadrilateral has angles 80° , 100° , 70° and 110° . A student asks whether copies of THIS quadrilateral can tile the plane. The key fact is:

- (A) No — only regular polygons tile, and this is irregular.
- (B) Yes — any quadrilateral tiles, because its four angles sum to 360° , and four copies (one of each angle) can be arranged to meet at every vertex.
- (C) No — its angles must each divide 360° for it to tile.
- (D) Only if it is a parallelogram or a square.

49. A vertex is to be closed using only regular polygons, with angles a triangle (60°), a square (90°), and TWO more equal regular polygons. Each of those two must have interior angle:

- (A) 105° , supplied by a regular polygon between a square and a hexagon.
- (B) 90° , two squares.
- (C) 120° , two hexagons.
- (D) 105° , which no regular polygon provides, so this configuration cannot be completed with two equal regular tiles.

50. Three regular polygons meet at a vertex: a triangle (60°), a regular dodecagon (150°), and one more. The third polygon's interior angle and name are:

- (A) 120° , a regular hexagon, since $360 - 60 - 150 = 150 \approx 120$.
- (B) 150° , another regular dodecagon, since $360 - 60 - 150 = 150^\circ$.
- (C) 90° , a square, since a triangle plus dodecagon usually leaves room for a square.
- (D) 150° , but no regular polygon has that angle.

51. A vertex uses one square (90°), one regular hexagon (120°), and the remainder filled by equilateral triangles (60°). How many triangles close it exactly?

- (A) Two triangles, since $360 - 90 - 120 = 150$ and $150 \div 60 \approx 2$.
- (B) Three triangles, since $3 \times 60 = 180$ fills the rest.
- (C) Two triangles, giving exactly 360° .
- (D) Two and a half, which is not whole, so squares-plus-hexagon-plus-triangles cannot close this particular vertex.

52. A regular hexagon (120°) and a regular dodecagon (150°) meet at a vertex with one more regular polygon. The leftover angle is 90° , but a student claims you could instead use TWO equal tiles there. Each would need:

- (A) 45° , supplied by half an octagon. (B) 30° , two of which make 60° .
- (C) 90° , two squares.
- (D) 45° , which no regular polygon provides, so two equal tiles cannot fill the 90° leftover; only one square works.

53. At a vertex one regular octagon (135°) is placed, and the rest filled with equilateral triangles (60°) only. Can the vertex close exactly?

- (A) Yes, with three triangles, since $3 \times 60 = 180 \approx 225$.
- (B) Yes, with four triangles, since $4 \times 60 = 240 \approx 225$.
- (C) Yes, with 3.75 triangles.
- (D) No, since $360 - 135 = 225$ and $225 \div 60 = 3.75$ is not a whole number.

54. A perpendicular bisector of segment GH is constructed, and a 60° angle is constructed at G on line GH. A student claims the 60° angle's far arm will always cross the perpendicular bisector. For a long enough arm, this is:

- (A) false, because the perpendicular bisector is vertical and the arm is horizontal.
- (B) true, because the perpendicular bisector and the slanted 60° arm are not parallel, so they must eventually meet.
- (C) false, because two lines from different points never meet.
- (D) true only if GH is shorter than 6 cm.

55. Six equilateral triangles tile around a point. If you remove one triangle and try to close the gap with a single regular polygon, that polygon needs interior angle:

- (A) 120° , a regular hexagon, since one hexagon replaces one triangle.
- (B) 300° , which is impossible for a regular polygon.
- (C) no polygon works, since the gap is irregular.
- (D) 60° , which is just another equilateral triangle — so nothing new is gained.

56. A right triangle is constructed by RHS with hypotenuse 25 cm and one leg 7 cm. The area of the triangle is:

- (A) 87.5 cm^2 , since $\text{area} = \frac{1}{2} \times 25 \times 7$. (B) 175 cm^2 , since $\text{area} = \frac{1}{2} \times 25 \times 14$.
- (C) It cannot be built, since 7 cm is too short.
- (D) 84 cm^2 , since the other leg is $\sqrt{(25^2 - 7^2)} = 24$ and $\text{area} = \frac{1}{2} \times 7 \times 24$.

57. A triangle is constructed by SAS with two sides 5 cm and 5 cm and included angle 90° . The triangle is:

- (A) an isosceles right triangle with hypotenuse $5\sqrt{2}$ cm.
- (B) equilateral with all sides 5 cm. (C) a right triangle with hypotenuse 10 cm.
- (D) impossible, since a right angle needs unequal sides.

58. You construct a 90° angle, bisect it to 45° , and then construct a 60° angle sharing one arm with the 45° , on the same side. The total angle is 105° . Bisecting THIS 105° angle would give 52.5° . Is 52.5° constructible this way?

- (A) No, since 52.5° is not a multiple of 15° .
- (B) No, since bisecting requires the angle to be even in degrees.
- (C) Yes, but only by trisecting 157.5° .
- (D) Yes, since $105^\circ = 60^\circ + 45^\circ$ is constructible and bisecting any constructible angle is allowed, giving 52.5° .

59. A semi-regular tiling repeats one vertex type. A candidate vertex lists triangle (60°), triangle (60°), triangle (60°), triangle (60°), triangle (60°). At this vertex the tiles total 360° , but a student says this makes it 'semi-regular'. The correct classification is:

- (A) It is semi-regular, because six tiles meet.
- (B) It is the ordinary (regular) triangular tiling, not semi-regular, since all tiles are the same regular polygon.
- (C) It is not a tiling, since six triangles overlap.
- (D) It is semi-regular, because the angles add to 360° .

60. The single rule that governs every interior vertex of any tiling — regular, semi-regular, or made of triangles and quadrilaterals — is that the corner angles meeting there sum to:

- (A) exactly 360° , the full turn — less leaves a gap, more forces an overlap.
- (B) exactly 180° , a straight angle. (C) at most 360° , so small gaps are allowed.
- (D) any multiple of 90° , such as 270° or 450° .

Solutions — Maths (Class 7), Chapter 14: Constructions and Tilings — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	D	31	C	41	D	51	D
2	C	12	A	22	B	32	A	42	C	52	D
3	C	13	C	23	B	33	D	43	B	53	D
4	D	14	B	24	C	34	B	44	C	54	B
5	A	15	A	25	A	35	A	45	A	55	D
6	C	16	A	26	C	36	C	46	B	56	D
7	A	17	A	27	D	37	D	47	B	57	A
8	A	18	D	28	B	38	C	48	B	58	D
9	B	19	C	29	A	39	D	49	D	59	B
10	B	20	C	30	C	40	B	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) A point equidistant from A and B lies on the perpendicular bisector of AB; equidistant from B and C means it lies on the perpendicular bisector of BC. The single point on both is their intersection, automatically equidistant from all three. Angle bisectors give the point equidistant from the SIDES (the incentre), not from the vertices. A midpoint of one segment is only equidistant from that segment's two ends. A whole perpendicular bisector handles only two towns, not all three, so a unique point is needed.

2. (C) Because $RP = RQ$, R lies on the perpendicular bisector of PQ, so $RM \perp PQ$ and M is the midpoint, giving $PM = 5$ cm. Triangle RMP is right-angled at M with hypotenuse $RP = 6$, so $RM^2 = 6^2 - 5^2 = 36 - 25 = 11$, $RM = \sqrt{11}$. Subtracting the lengths $(6 - 5)$ ignores the right angle. The radius 6 is the slant RP, not the vertical RM. Adding the squares would apply if RM were the hypotenuse, but here RP is.

3. (C) Triangle FNK has a right angle at N, so FK is its hypotenuse and FN is a leg; the hypotenuse is always the longest side, hence $FN < FK$. The perpendicular distance is the

unique shortest distance from a point to a line, not equal to a slanted one. It is shortest, not longest, so $FK < FN$ is reversed. The right-triangle inequality settles the comparison without needing NK 's length.

4. (D) The centre is equidistant from A and B, so it lies on the perpendicular bisector of AB, whose foot is the midpoint, splitting the chord into 6 cm halves. The radius OA is the hypotenuse of a right triangle with legs 6 (half-chord) and 8 (distance to chord): $\text{radius}^2 = 6^2 + 8^2 = 36 + 64 = 100$, radius = 10. The 8 cm is only a leg, not the radius. Adding $6 + 8$ ignores the right-angle relation. Subtracting the squares treats the half-chord as the hypotenuse, which it is not.

5. (A) W lies on the bisectors of two sides, so it is equidistant from all three vertices; that makes it equidistant from the third side's two endpoints, so it lies on that side's perpendicular bisector too. Hence all three are concurrent at the circumcentre for ANY triangle, not only equilateral ones. The 'three lines rarely meet' intuition fails here because of the shared equidistance property. The circumcentre is generally not on any side, so it is not that side's midpoint.

6. (C) The perpendicular bisector meets AB at the midpoint, so half of AB is 12 cm. Triangle with hypotenuse TA = 13 and one leg 12 gives the other leg = $\sqrt{(13^2 - 12^2)} = \sqrt{(169 - 144)} = \sqrt{25} = 5$ cm, the distance from T down to AB. The slant TA (13) is not the perpendicular distance. Subtracting lengths $(13 - 12)$ ignores Pythagoras. Adding the squares treats TA as a leg, but it is the hypotenuse.

7. (A) Since $AL = AR$, A is the midpoint of LR. T, found by equal arcs from L and R, is equidistant from them, so it lies on the perpendicular bisector of LR — which passes through midpoint A and is perpendicular to LR (i.e. m). Thus AT is both the perpendicular bisector and perpendicular to m, so both claims hold. The radius only needs to exceed half of LR for the arcs to cross; it need not equal AL.

8. (A) Bisecting 60° gives 30° ; bisecting that gives 15° (two halvings, so $60 \div 4$). Adding 15° to the original 60° gives 75° . Three halvings would give 7.5° , but only two are described. The single half (30°) added gives 90° , but the question takes the half OF the half. A 20° piece would require trisecting 60° , which is impossible with compass and straightedge, so 80° is unreachable here.

9. (B) $50^\circ = 150^\circ \div 3$, and $150^\circ (= 60^\circ + 90^\circ)$ is constructible, but trisecting a general angle with these tools is impossible, so 50° is not constructible. By contrast $75^\circ = 60^\circ + 15^\circ$ where $15^\circ = \text{half of } 30^\circ = \text{half of half of } 60^\circ$, all bisections. $105^\circ = 60^\circ + 45^\circ$ with $45^\circ = \text{half of a constructed right angle}$. $7.5^\circ = 60^\circ$ halved three times. Each of those uses only bisection and addition of constructible angles, so they ARE constructible.

10. (B) $67.5^\circ = 45^\circ + 22.5^\circ$. A right angle (90°) is constructible; halving gives 45° , halving again gives 22.5° ; adding $45^\circ + 22.5^\circ$ gives 67.5° — all by bisection and addition. Trisecting

a general angle is impossible, so the 202.5° route fails. The $135^\circ = 90^\circ + 45^\circ$ is itself constructible, and halving it does give 67.5° , BUT that distractor justifies 135° by trisecting 405° , an impossible step, so it is invalid as stated. Trisecting 22.5° to reach 7.5° is also impossible.

11. (C) Starting from 90° : first halving 45° , second 22.5° , third 11.25° . Three successive bisections of 90° give $90 \div 8 = 11.25^\circ$. Dividing 90 by 12 is not a bisection operation. Subtracting 7.5° is not part of any bisection chain. 5.625° would require a fourth halving, one more than the three stated.

12. (A) Three 30° angles total $3 \times 30 = 90^\circ$. Bisecting that 90° gives 45° . Bisecting does not 'return' a single 30° piece — it halves the whole built-up angle. Subtracting 30° is not what bisecting does. No 45° is added; the operation is a single bisection, giving 45° , not 135° .

13. (C) Halving the GIVEN 96° angle is allowed regardless of whether 24° is constructible from scratch; $96 \div 2 = 48$ and $48 \div 2 = 24$, two valid bisections. The 'multiple of 15° ' rule is about building from $60^\circ/90^\circ$, not about halving a supplied angle, so it does not block this. Trisection of a general angle is impossible, so the 32° route fails. Halving once gives 48° , which is not 24° — closeness does not count.

14. (B) An angle is fixed by the ratio of chord to radius, not the chord alone. Keeping chord PQ the same while shrinking the radius means the chord now spans a larger fraction of a smaller circle, so it subtends a wider angle. Copying the chord faithfully only reproduces the angle when the SAME radius is used. A smaller radius does not directly mean a smaller angle; with a fixed chord it means a larger angle. The outcome is predictable from the chord-to-radius reasoning, no measuring needed.

15. (A) A long length is reproduced by stepping off smaller compass-carried pieces end to end along a straight line, each transferred exactly, so $7 = 4 + 3$ works within a 5 cm span. The claim that it 'can never be copied' ignores stepping off. A single 7 cm arc is impossible when the span is only 5 cm. A marked ruler is measuring, not constructing; the compass plus straightedge suffice via stepping off.

16. (A) The triangle inequality gives $9 - 4 < x < 9 + 4$, i.e. $5 < x < 13$. The integers strictly between are 6, 7, 8, 9, 10, 11, 12 — exactly 7 values. Including the endpoints 5 and 13 (giving 9 values) is wrong, since those degenerate into straight lines. The difference $9 - 4 = 5$ is just the lower bound, not a count. Counting from 5 up to 12 wrongly includes 5, which is excluded.

17. (A) The three angles of a triangle sum to 180° , so two of them must total under 180° to leave a positive third. Here $95^\circ + 86^\circ = 181^\circ > 180^\circ$, so the arms from the base ends diverge and never meet; no triangle exists. A negative angle is impossible. A positive base

cannot rescue an impossible angle pair. ASA does not require equal base angles, only that their sum stays below 180° .

18. (D) When the included angle is 90° , the two given sides are the legs and the opposite side is the hypotenuse: $\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$. Adding the sides (14) or subtracting them (2) ignores the right-angle (Pythagorean) relationship. Subtracting the squares would apply only if one given side were the hypotenuse, but here both are legs.

19. (C) Two sides with the angle BETWEEN them is the SAS condition, which fixes a unique triangle. An angle OPPOSITE one of the sides is the ambiguous SSA case, which can give two non-congruent triangles. Three angles fix the shape but not the size, so infinitely many triangles fit. Sides 5, 6 and 11 fail the triangle inequality ($5 + 6 = 11$ is not greater than 11), so no triangle forms at all.

20. (C) The missing leg satisfies $\text{leg}^2 + 8^2 = 17^2$, so $\text{leg}^2 = 289 - 64 = 225$, $\text{leg} = 15$. Perimeter = $8 + 15 + 17 = 40$ cm. Taking the leg as $17 - 8 = 9$ ignores Pythagoras (and the arithmetic given is inconsistent). Taking it as $17 + 8 = 25$ also ignores the right-angle relation. RHS requires the hypotenuse to be the longest side, not equal to a leg, so the triangle is perfectly constructible.

21. (D) With $AB = AC$, the base angles are equal; since they sum to $180 - 60 = 120^\circ$, each is 60° . All three angles are 60° , so the triangle is equilateral and $BC = 6$ cm. It is not right-angled (no 90° angle appears). It is not merely isosceles-with-longer-base; the third side equals the others. SAS happily allows two equal sides; that is the standard isosceles/equilateral case.

22. (B) With two sides hinged at the included angle, opening the angle wider pushes the far endpoints further apart, so the opposite side grows monotonically as the angle increases from 30° toward 150° . Fixing the two sides does NOT fix the third — it depends on the angle. A wider angle spreads, not folds, the endpoints. The growth is continuous up to 180° , with no peak at 90° .

23. (B) The side opposite the 90° angle is the hypotenuse, and the other two (3 and 4) are the legs: $\text{hypotenuse} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$. The sum 7 or difference 1 ignores the Pythagorean relation. A right angle opposite a specified side, with the other two sides given, fixes the hypotenuse exactly at 5, so it is not free to be any length over 4.

24. (C) Exterior angle = $180 - 168 = 12^\circ$. The exterior angles of a regular n -gon sum to 360° , so $n = 360 \div 12 = 30$. Dividing 168 by an arbitrary 14 has no geometric basis. Using $180 \div \text{exterior}$ instead of $360 \div \text{exterior}$ is the wrong total. Dividing 168 by 8 is also meaningless for finding sides.

25. (A) More sides means a larger interior angle, so $120 + 30 = 150^\circ$. Exterior = $180 - 150 = 30^\circ$, and $n = 360 \div 30 = 12$, a regular dodecagon. The octagon's interior angle is 135° ,

not 150° . A decagon's interior angle is 144° , not 150° . The figure $360 \div 150 = 2.4$ counts tiles at a vertex, not sides, and rounding it to 9 is meaningless.

26. (C) Interior + exterior = 180° , and interior = $5 \times$ exterior, so $6 \times$ exterior = 180° , exterior = 30° . Then $n = 360 \div 30 = 12$, a dodecagon. 'One-fifth' is a ratio of angles, not a count of sides, so neither 'hexagon' nor 'pentagon' follows. For a decagon exterior = 36° and interior = 144° , and $144 \div 5 = 28.8 \neq 36$, so the fifth relation fails there.

27. (D) If interior = $2 \times$ exterior and they sum to 180° , then $3 \times$ exterior = 180° , exterior = 60° , so $n = 360 \div 60 = 6$ (the hexagon: 120° interior, 60° exterior, and $120 = 2 \times 60$). For a square, interior 90° and exterior 90° are equal, not double. For a triangle, interior 60° and exterior 120° , so interior is HALF the exterior, not double. 'n = 12 because 12 is double 6' confuses side count with the angle ratio.

28. (B) If $360 \div$ interior = 2.5, then interior = $360 \div 2.5 = 144^\circ$. Exterior = $180 - 144 = 36^\circ$, so $n = 360 \div 36 = 10$, a decagon. For a dodecagon $360 \div 150 = 2.4$, not 2.5. For an octagon $360 \div 135 \approx 2.67$. For a square $360 \div 90 = 4$. Only the decagon gives exactly 2.5, confirming it cannot self-tile (2.5 is not whole).

29. (A) The self-tilers are the triangle, square and hexagon. Their EXTERIOR angles are $180 - 60 = 120^\circ$, $180 - 90 = 90^\circ$, and $180 - 120 = 60^\circ$ respectively. The list 60° , 90° , 120° is the INTERIOR angles, not exterior. The pentagon (72° exterior) and octagon (45° exterior) do not self-tile. Including the pentagon (72° exterior) is wrong because the pentagon does not tile.

30. (C) As n grows past 6, the interior angle rises above the hexagon's 120° but stays below 180° . So $360 \div$ interior falls strictly between $360 \div 180 = 2$ and $360 \div 120 = 3$, a non-integer, so no whole number of corners can close a vertex. Interior angles never reach 180° for a real polygon, so 'exceeds 180° ' is false. Side count alone does not block tiling — the angle-divisibility does. The exterior angle ($360/n$) stays positive, never negative.

31. (C) Self-tiling needs $k \times$ (interior angle) = 360° with interior = $180 - 360/n$ a valid regular-polygon angle. The only solutions are the triangle ($k = 6$), square ($k = 4$) and hexagon ($k = 3$), so $k \in \{3, 4, 6\}$. $k = 5$ would need a 72° interior angle, which no regular polygon has. $k = 2$ would need a 180° angle, impossible. Not every $k \geq 3$ works — $k = 5$ is the clearest gap.

32. (A) Self-tiling requires $360 \div$ interior to be a whole number; $360 \div 150 = 2.4$ is not, so it fails. Two corners give $2 \times 150 = 300^\circ$, a 60° gap; three give 450° , a 90° overlap. Rounding the divisor up or down does not fix the gap or overlap. Being under 180° is necessary but far from sufficient — exact divisibility is the rule.

33. (D) Self-tiling needs $360 \div$ interior to be a positive integer. Here $360 \div 108 \approx 3.33$, $360 \div 135 \approx 2.67$, $360 \div 144 = 2.5$ — none integers — so none self-tile. Being a multiple of

45° does not help the octagon (2.67 is still fractional). Exceeding 120° is irrelevant. Under- 180° is necessary but not sufficient; all three fail the exact-division test.

34. (B) For $90^\circ < A < 120^\circ$, dividing 360 gives $360 \div 120 = 3 < 360 \div A < 360 \div 90 = 4$, so the quotient is strictly between 3 and 4 — never a whole number, so no self-tiling. The pentagon's 108° is in this band and does NOT tile ($360 \div 108 \approx 3.33$), and it is not a multiple of 30° anyway. 'Three or four can fit' is exactly the rounding trap. A in this band is well under 180° , so that claim is false.

35. (A) A valid interior vertex must total exactly 360° . A total of 354° is 6° short, so an uncovered wedge (gap) remains and the vertex is invalid. A shortfall is a gap, not an overlap (an overlap comes from totals above 360°). 'Within rounding distance' is the classic trap; only exact 360° works. Totals under 360° are entirely possible — they simply signal gaps.

36. (C) Summing: $4 \times 60 = 240^\circ$ plus $2 \times 90 = 180^\circ$ gives 420° , which exceeds 360° by 60° , so the tiles pile up — an overlap. The number of tiles (six) does not guarantee closure; the angle sum does. Exceeding 360° is an overlap, not a gap. That the individual angles divide 360° is irrelevant; only their sum at the vertex matters.

37. (D) Add the corners: $90 + 90 + 60 + 60 = 300^\circ$, which is 60° short of 360° , so a gap remains and the vertex does NOT close. A sum of 300° is not 360° . A shortfall is a gap, not an overlap (an overlap would need a total over 360°). The count of four tiles is incidental; the sum 300° is what decides, and it fails.

38. (C) The leftover angle is $360 - 120 - 90 = 150^\circ$, and a 150° interior angle belongs to the regular dodecagon (12 sides). A hexagon (120°) would give $120 + 90 + 120 = 330^\circ$, a gap. A square (90°) gives 300° , a bigger gap. A triangle (60°) gives 270° , an even bigger gap. Only 150° closes the vertex exactly; no rounding is allowed.

39. (D) The leftover is $360 - 150 - 120 = 90^\circ$, which is exactly a square's interior angle, so a square closes the vertex (this is the 3.4.3.12-family arithmetic; the third tile is a square). A triangle (60°) would leave a 30° gap. A second hexagon (120°) would overshoot. A 90° regular polygon certainly exists — it is the square — so the 'no such polygon' claim is false.

40. (B) Leftover = $360 - 135 - 90 = 135^\circ$, which is the octagon's own interior angle, so a second octagon closes the vertex (the familiar square-and-octagon tiling has two octagons and one square at each vertex). A square (90°) would total only 315° . A hexagon (120°) would total 345° . A 135° regular polygon does exist — it is the octagon — so the 'no such polygon' claim is false.

41. (D) Leftover = $360 - 150 = 210^\circ$, and $210 \div 60 = 3.5$, not a whole number, so triangles alone cannot fill it (you would need 3.5 tiles). Three triangles give 180° , totalling 330° (a gap). Four triangles give 240° , totalling 390° (an overlap). Neither three nor four closes it, so the configuration fails with triangles alone.

42. (C) Leftover = $360 - 120 = 240^\circ$, and each triangle is 60° , so $240 \div 60 = 4$ triangles close it exactly. Dividing 240 by 120 wrongly uses the hexagon's angle for triangles. Three triangles give only 180° , leaving a 60° gap. Six triangles (360°) would leave no room for the hexagon, overshooting.

43. (B) Two hexagons use $2 \times 120 = 240^\circ$, leaving $360 - 240 = 120^\circ$. Filling 120° with squares needs $120 \div 90 = 1.33$ squares, not whole, so no exact closure. One square gives 90° , totalling 330° (a 30° gap). Two squares give 180° , totalling 420° (an overlap). A fractional 1.33 squares is impossible, so it cannot close.

44. (C) Summing the knowns: $60 + 60 + 90 = 210^\circ$, leaving $360 - 210 = 150^\circ$, which is the dodecagon's interior angle (the 3.3.4.12 vertex). A hexagon (120°) would total 330° , a gap. A second square would total 300° , a bigger gap. A 150° regular polygon does exist (the dodecagon), so the 'no such polygon' claim is false.

45. (A) Add the corners: $90 + 3 \times 60 = 90 + 180 = 270^\circ$, which is 90° short of 360° , so a gap remains and these four tiles do NOT close the vertex (you would need, for example, one more square to reach 360° , giving the 3.3.3.4.4 arrangement). $90 + 180 = 270$, not 360. A shortfall is a gap, not an overlap. The count of four tiles is incidental; the 270° sum decides, and it fails.

46. (B) The leftover is $360 - 108 = 252^\circ$. For k equal tiles of interior angle A to fill it, $kA = 252$ with A a real regular-polygon angle. Testing triangles ($252 \div 60 = 4.2$), squares ($252 \div 90 = 2.8$), hexagons ($252 \div 120 = 2.1$) all give non-integers, and 252° itself is impossible (over 180°). So no single repeated regular polygon closes a pentagon vertex — this is why the pentagon resists semi-regular tilings with one companion. The 126° idea misreads the leftover as 126 instead of 252. Rounding 4.2 to 4 or 2.8 to 3 leaves a gap or overlap.

47. (B) Rotating a copy of any triangle 180° about a side's midpoint forms a parallelogram, which tiles the plane by translation, so EVERY triangle tiles. At each vertex the three different angles (summing to 180° , so two full sets give 360°) arrange to close exactly. The angles-divide- 360° rule is for a single REGULAR polygon meeting copies of itself corner-to-corner, not for general triangles. Both acute and obtuse triangles tile this way.

48. (B) The angles of any quadrilateral sum to 360° . By rotating copies, all four different angles can be brought to meet at each vertex, summing exactly to 360° , so EVERY quadrilateral (even non-convex) tiles the plane. The 'only regular polygons tile' rule applies to a single regular shape meeting itself, not to clever arrangements of an irregular tile. Individual angles need not divide 360° ; it is their full set summing to 360° that matters. Parallelograms and squares are just special cases.

49. (D) Leftover after the triangle and square is $360 - 60 - 90 = 210^\circ$, split equally between two tiles, so each needs 105° . The regular-polygon interior angles go 60° , 90° ,

108° , 120° , ... — none is 105° , so no such tile exists and the vertex cannot be completed this way. Two squares (90° each) give $60 + 90 + 90 + 90 = 330^\circ$ (gap). Two hexagons give $60 + 90 + 120 + 120 = 390^\circ$ (overlap). The needed 105° simply does not occur.

50. (B) Leftover = $360 - 60 - 150 = 150^\circ$, the dodecagon's interior angle, so a second dodecagon closes the vertex (the 3.12.12 tiling). A hexagon (120°) would total 330° , a gap. A square (90°) would total 300° , a bigger gap. A 150° regular polygon exists — the dodecagon — so the 'no such polygon' claim is false.

51. (D) Leftover = $360 - 90 - 120 = 150^\circ$, and $150 \div 60 = 2.5$, not whole, so triangles cannot exactly fill it (2.5 tiles is impossible). Two triangles give 120° , totalling 330° (a 30° gap). Three triangles give 180° , totalling 390° (an overlap). Neither closes it, so this combination fails.

52. (D) Leftover = $360 - 120 - 150 = 90^\circ$. Splitting 90° between two equal regular tiles needs 45° each, but the smallest regular interior angle is 60° (the triangle), so no 45° regular polygon exists. 'Half an octagon' is not a regular tile. Two 30° tiles would give only 60° , not 90° . Two squares (90° each) would give 180° , far over the 90° leftover. So a single square (90°) is the only fit.

53. (D) Leftover = $360 - 135 = 225^\circ$, and $225 \div 60 = 3.75$, not whole, so triangles alone cannot fill it. Three triangles give 180° , totalling 315° (a gap). Four triangles give 240° , totalling 375° (an overlap). A fractional 3.75 triangles is impossible. So an octagon plus triangles alone cannot close the vertex.

54. (B) The perpendicular bisector is perpendicular to GH; the 60° arm makes a 60° angle with GH, so the two are not parallel (their angles with GH differ). Two non-parallel lines in a plane always intersect when extended, so a long enough arm crosses the bisector. The arm is not 'horizontal' — it tilts at 60° . Lines from different points certainly can meet. The length of GH does not affect whether two non-parallel lines eventually cross.

55. (D) Five triangles cover $5 \times 60 = 300^\circ$, leaving $360 - 300 = 60^\circ$. A 60° interior angle is exactly an equilateral triangle, so the only regular fit is another triangle (restoring the original six). A hexagon (120°) would overshoot the 60° gap. The leftover is 60° , not 300° . A 60° gap is filled perfectly by a triangle, so a regular polygon does work.

56. (D) The missing leg is $\sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. The two legs (7 and 24) are perpendicular, so area = $\frac{1}{2} \times 7 \times 24 = 84 \text{ cm}^2$. Using the hypotenuse (25) as a base with leg 7 ignores that the hypotenuse is not perpendicular to a leg. The leg 7 is fine — RHS only needs the hypotenuse to be the longest side, which 25 is.

57. (A) Two equal legs with a 90° included angle give an isosceles right triangle; the hypotenuse is $\sqrt{5^2 + 5^2} = \sqrt{50} = 5\sqrt{2}$. It is not equilateral, since the hypotenuse ($5\sqrt{2} \approx 7.07$) exceeds the legs. The hypotenuse is $5\sqrt{2}$, not 10 (10 would need legs summing in a

way Pythagoras forbids). A right angle can perfectly well sit between two equal legs, so it is not impossible.

58. (D) 105° is built from constructible parts ($60^\circ + 45^\circ$), and bisecting any already-constructed angle is a valid compass-straightedge step, yielding 52.5° . The 'multiple of 15° ' idea is not a real constructibility rule — bisection halves whatever you have. Angles need not be 'even in degrees' to be bisected; the compass halves them geometrically. Trisection of a general angle is impossible, so that route is invalid anyway.

59. (B) Six triangles each contributing 60° total exactly 360° , so the vertex closes — but a semi-regular tiling needs TWO OR MORE different regular polygons. With only triangles it is the regular (monohedral) triangular tiling, not semi-regular. Six triangles close exactly with no overlap. Adding to 360° makes it a valid tiling, but sameness of tile keeps it regular, not semi-regular.

60. (A) Around any interior point there are 360° of turning, so the tile corners meeting there must total exactly 360° — under it leaves an uncovered wedge (gap), over it makes tiles overlap. 180° is only a straight angle along an edge, not a vertex full turn. 'At most 360° ' permits gaps, which violate the tiling definition. Multiples of 90° like 270° (gap) or 450° (overlap) do not close a vertex; only 360° does.